

Malvika Raj Joshi

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Research Interests: Quantum computing, quantum-classical separations, privacy & cryptography.

Education

University of California, Berkeley 2021–Present

Ph.D. in Computer Science. Advisors: Umesh Vazirani & John Wright

Research area: Quantum complexity theory.

Massachusetts Institute of Technology 2020–2021

M.Eng. in EECS. Advisor: Aram Harrow.

Thesis: *Pretending to be Quantum: A Study of IQP-Based Tests of Quantumness*

Massachusetts Institute of Technology 2016–2020

B.S. in Physics and B.S. in Computer Science. GPA: 5.0/5.0.

Publications

- Lucas Greta, Meghal Gupta, and **Malvika Raj Joshi**. *Parity $\notin$ QAC⁰ $\iff$ QAC⁰ is Fourier-Concentrated*. (FOCS, 2026).
- Lucas Greta and **Malvika Raj Joshi**. *Shor's Algorithm Requires Fanout*, 2026. arXiv:2608.06703.
- Lucas Greta, Meghal Gupta, and **Malvika Raj Joshi**. *Polylogarithmic-Weight Dicke States in QAC⁰ and Arbitrary Symmetric States in QAC_f⁰*, 2026. arXiv:2604.15298.
- **Malvika Raj Joshi**, Avishay Tal, Francisca Vasconcelos, and John Wright. *Improved Lower Bounds for QAC⁰*. (STOC, 2026).
- **Malvika Raj Joshi** and Francisca Vasconcelos. *Constant-Depth Unitary Preparation of Dicke States*, 2026. arXiv:2601.10693.
- Shyan Akmal, Lijie Chen, Ce Jin, **Malvika Raj**, and Ryan Williams. *Improved Merlin–Arthur Protocols for Central Problems in Fine-Grained Complexity*. (ITCS, 2022).

Industry Experience

Amazon, East Palo Alto — *Applied Scientist Intern* May–Aug. 2026

Differentially-Private Diffusion with improved utility. Preprint in preparation.

Amazon, East Palo Alto — *Applied Scientist Intern* May–Aug. 2025

Designed *Lexicure*, a leakage-free encrypted lexical search protocol, and developed new secure primitives for higher-degree arithmetic. *With:* Tal Wagner, Shai Halevi, Nina Mishra.

Facebook, New York, NY — *Software Engineering Intern (PyTorch)* May–Aug. 2019

Designed and implemented a C++ API for PyTorch Autograd introducing native C++ support for automatic differentiation.

Facebook, Menlo Park, CA — *Software Engineering Intern* May–Aug. 2018

Optimized the News Feed ranking pipeline by parallelizing components in C++.

WhatsApp, Menlo Park, CA — *Software Engineering Intern* Jun.–Sep. 2017

Designed and implemented a caching mechanism for encrypted downloads in backend infrastructure.

Honors and Awards

- **International Olympiad in Informatics (IOI)**: Silver (2014, 2015; ranks 28, 29); Bronze (2016).
- **Asia Pacific Informatics Olympiad (APIO)**: Silver (2014); Bronze (2015, 2016).
- Elected member of **Sigma Pi Sigma** (Physics Honor Society).
- Microsoft Q# Coding Contest 2019: Top 50.

Outreach and Professional Activities

INOI Training Initiative for Women

2020–2021

Founder and Volunteer

Organized and taught algorithms and programming to female high school students preparing for the Indian National Olympiad in Informatics (INOI). Developed curriculum, delivered lectures, and provided mentorship.

Massachusetts Institute of Technology

June–Aug. 2020

Research Intern, CSAIL (Advisor: Dina Katabi)

Processed large-scale video and RF data for sleep monitoring, designed and implemented caching mechanism for Unicorn-based server.

Competitive Programming

2015–2017

Problem Setter

Problem setter for programming contests including Indian IOI selection rounds (IOITC, ZIO/ZCO), ACM ICPC Chennai Regionals, and CodeChef Long Challenges.

Massachusetts Institute of Technology

Oct.–Dec. 2016

UROP Researcher (Advisor: Nancy Kanwisher)

Developed components of an iOS application for human eye-tracking experiment.

Teaching Experience

University of California, Berkeley

Graduate Student Instructor (CS70 Fall 2021, CS170 Fall 2023, CS270 Fall 2025)

Led discussions, office hours, and graded coursework; additionally created homeworks and exams (CS170).

Massachusetts Institute of Technology

Grader, 6.045: Automata, Computability, and Complexity Theory (Sp. 2018, 2020)

Skills

Codeforces International Master (peak 2318); **CodeChef** 7-star (peak 2552); **Topcoder** top 4%.

Programming languages: C, C++, Python, Java, Assembly.